

## **SELF■CALIBRATION, DIAGNOSTICS, AND PERFORMANCE VERIFICATION EMBODIMENTS**

In various embodiments, the intraluminal compliance and distance profiling device incorporates automated self■calibration and diagnostic systems configured to maintain measurement accuracy, detect performance degradation, and verify operational readiness prior to or during measurement sessions.

The device may perform initialization diagnostics upon activation, including verification of sensor functionality, pressure response validation, communication integrity testing, battery health assessment, and confirmation of memory availability. Initialization checks may prevent measurement initiation when system readiness criteria are not satisfied.

In certain embodiments, distance sensing components may be calibrated using internal reference geometries, known reflective targets, or calibration fixtures integrated within docking accessories. Calibration routines may compensate for sensor drift, temperature variation, or manufacturing tolerances affecting measurement precision.

For inflatable compliance embodiments, calibration procedures may verify pressure sensor accuracy, valve response behavior, and inflation system sealing integrity. Controlled test inflations using known internal volumes may confirm expected pressure response prior to user operation.

The system may continuously monitor diagnostic indicators during operation. Real■time diagnostics may evaluate signal stability, noise levels, sensor saturation conditions, communication latency, or unexpected mechanical behavior. Diagnostic monitoring may allow automatic correction or termination of unreliable measurements.

In some implementations, predictive maintenance algorithms may analyze historical device performance trends to anticipate component wear or failure. Users may receive maintenance notifications or recommended service intervals before performance degradation affects measurement reliability.

Diagnostic information may be communicated to users through simplified status indicators or detailed engineering reports depending on operational mode. Consumer deployments may receive intuitive readiness indicators, while research or service modes may expose advanced diagnostic metrics.

In certain embodiments, remote diagnostic support may be enabled through secure transmission of anonymized performance logs to authorized service platforms. Remote analysis may assist troubleshooting, firmware optimization, or validation of field performance across deployed devices.

Self■calibration and diagnostic systems may further support cross■device consistency by aligning measurement outputs across multiple devices or product generations using standardized calibration references and computational normalization techniques.

Accordingly, self■calibration and performance verification embodiments enhance reliability, extend device lifespan, reduce measurement variability, and ensure sustained analytic validity throughout long■term deployment of the disclosed physiologic measurement system.